

Do Supervised and Unsupervised Methods Agree on BRCA Subtypes?

DATS 6450, Multi-omic TCGA-BRCA at scale

Aidan Carlisle

Zachary Cardell

May 5, 2026

Abstract We ingest six TCGA-BRCA molecular modalities (RNA-seq, methylation, CNV, RPPA, miRNA, mutations) directly from a public S3 bucket, with 444 million rows of methylation alone (exceeding single-node memory), and ask whether **supervised** (XGBoost on PAM50 labels) and **unsupervised** (MOFA+ multi-omic factor analysis) approaches recover the same molecular structure on the integrated 6-view dataset. Both succeed and converge. XGBoost reaches 0.87 ± 0.01 5-fold-CV accuracy on PAM50, with 17 of 20 panel genes recovered in its top-50 SHAP features. MOFA+, given no labels, produces 14 active factors whose dominant factor loads jointly on all six modalities, and whose downstream k-means cluster is **97% pure Basal-like**. The two methods independently surface the same basal-vs-luminal axis, confirming it as the dominant structural signal in the data.

Table of contents

1 Introduction	2
1.1 Problem	2
1.2 Ten project questions	2
2 Dataset and Infrastructure	3
3 Exploratory Data Analysis	3
3.1 Q1: Multi-omic cohort funnel	3
3.2 Q2: Methylation missingness, technical with consequences	4
3.3 Q3: Most-recurrent BRCA mutations are textbook drivers	4
3.4 Q4: Methylation breaks single-node tools	4
3.5 Q5: Does the merged feature space preserve modality-specific signal?	5
4 Supervised: XGBoost on PAM50	5
4.1 Q6: Multi-omic XGBoost predicts PAM50 well	5
4.2 Q7: Top SHAP features recover the PAM50 panel	5
5 Unsupervised: MOFA+ on Six Modalities	6

5.1 Q8: MOFA+ identifies a multi-omic dominant factor	6
5.2 Q9: Clustering on factor scores partially recovers PAM50	8
5.3 Q10: Supervised and unsupervised methods converge on the same axis	9
6 Discussion	10
6.1 Synthesis	10
6.2 Limitations	10
6.3 Future work	11
7 References	11
8 Appendix	11
8.1 A. Methods detail	11
8.2 B. Cross-cohort sanity	11
8.3 C. Reproducibility	12
Bibliography	12

1 Introduction

1.1 Problem

On a multi-omic TCGA-BRCA dataset spanning six modalities and ~744 patients, do supervised and unsupervised analyses recover the same molecular-subtype structure?

Breast cancer (BRCA) is heterogeneous. The **PAM50** transcriptional panel [1] divides it into five intrinsic subtypes (Basal, Her2, LumA, LumB, Normal-like) using only RNA-seq. PAM50 is the de facto label, but BRCA biology spans modalities PAM50 does not see: methylation, copy number, protein abundance, microRNA, somatic mutations. We ask whether a multi-omic representation either (a) lets a supervised classifier sharpen PAM50 prediction, or (b) lets an unsupervised model recover PAM50-like structure without ever seeing a label, and whether the two methods agree on what the dominant axis of variation is.

1.2 Ten project questions

We organize the analysis around 10 specific questions.

Exploratory (Q1–Q5): feasibility, data quality, and feature-space sanity.

1. How does TCGA modality coverage funnel down to a multi-omic BRCA cohort?
2. Is methylation missingness technical or biological, and how does it interact with feature selection?
3. Are the most-recurrent BRCA mutations known cancer drivers (sanity)?
4. How does per-modality data volume scale, and where does single-node compute fail?
5. Does the merged six-view feature space preserve modality-specific structure or average it out?

Supervised ML (Q6–Q7): XGBoost on PAM50 with SHAP attribution.

6. Can XGBoost predict PAM50 subtypes from multi-omic features, and does it generalize to other cohorts?

7. Do the model’s top SHAP features match the published PAM50 marker panel?

Unsupervised ML (Q8–Q10): MOFA+ multi-omic factor analysis.

8. Does MOFA+ identify shared structure across all six modalities?

9. Does k-means clustering on those factors recover PAM50 subtypes?

10. Do the supervised and unsupervised methods converge on the same dominant biological axis?

2 Dataset and Infrastructure

All data originate from the public TCGA mirror at `s3://tcga-2-open/`. The six BRCA modalities used here are summarized in Table 1. Methylation is the bottleneck modality: **444 million rows / 5.5 GB compressed**, ~30 GB decompressed, exceeding the 7.6 GB single-node memory budget if loaded eagerly.

Table 1: BRCA modality volumes after ingest. Methylation crosses the ‘big-data’ threshold.

	Modality		Schema	Rows	Compressed
0	RNA-seq	gene × patient	FPKM	74.7 M	1.6 GB
1	methylation	probe × patient	β -value	444 M	5.5 GB
2	CNV	segment × patient	log2	463 K	7 MB
3	RPPA	peptide × patient		443 K	3.5 MB
4	miRNA	miRNA × patient	RPM	2.27 M	7 MB
5	mutations	gene × patient	(0/1)	65 K	0.4 MB

Compute is a single AWS EC2 t3 (2 vCPU, 7.6 GB RAM, no swap). To stay within memory, we use **DuckDB streaming** for any aggregation whose intermediate hash map is proportional to source rows (variance ranking on the methylation parquet) and **Polars lazy** scans (`scan_parquet().filter().select().collect(engine="streaming")`) for filter-and-project workloads. Eager `pl.read_parquet` on the methylation parquet reliably OOM-kills the instance, verified three times in development. A multi-node Spark-on-EC2 cluster is scaffolded but not on the critical path for the BRCA results below; the streaming patterns proved sufficient. Per-stage methods detail is in Appendix A.

The pipeline has four stages: ingest from S3 (per-modality readers in `src/ingest*_polars.py`), wide-format merge with per-view feature pre-selection (`src/merge.py`), supervised classification (`code/run_classification_pipeline.py`), and unsupervised factor analysis (`code/run_mofa.py` and `code/analyze_mofa.py`).

3 Exploratory Data Analysis

3.1 Q1: Multi-omic cohort funnel

The 6-way intersection collapses 800 to 1,100 per-modality patients down to **744 multi-omic patients**, the analysis cohort for everything that follows.

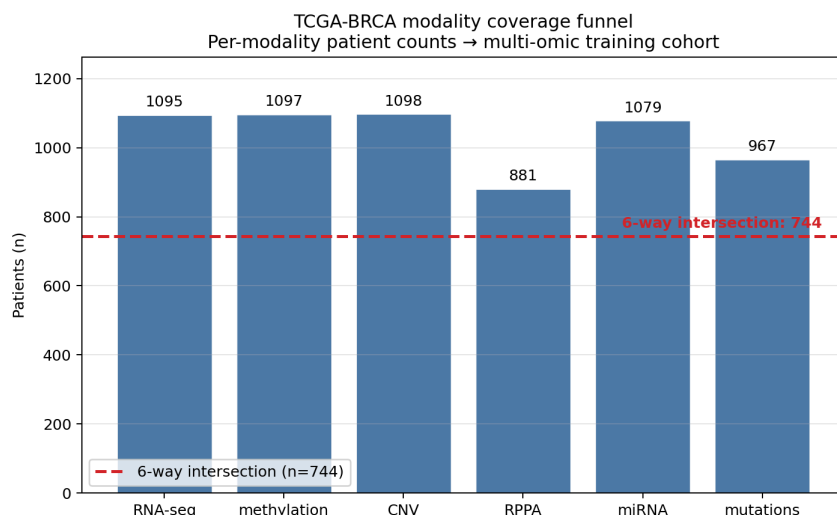


Figure 1: BRCA per-modality patient counts. The 6-way intersection (744 patients) defines the multi-omic training cohort.

3.2 Q2: Methylation missingness, technical with consequences

Methylation missingness is bimodal. Most probes sit at the array's ~18% baseline QC rate, but a long tail is **100% missing for TCGA-BRCA** (technical failures). A naive variance ranking selected those tail probes; adding a coverage gate ($\text{HAVING COUNT}(\ast) \geq 80\% \times n_patients$) dropped merged 6-view methylation missingness from **36.3% to 26.7%**, with the per-column p90 falling below 30%.

The lesson: any feature-selection statistic computed on the long-format parquet must be coverage-gated *first*, not variance-ranked first and filtered after. Otherwise a probe with two non-null β -values produces huge “empirical variance” and dominates the top-N.

3.3 Q3: Most-recurrent BRCA mutations are textbook drivers

The five most recurrent BRCA mutations are TP53, PIK3CA, TTN, GATA3, and CDH1, all classical BRCA drivers, which serves as a sanity check on the mutation ingest.

The 2%-recurrence filter in `src/merge.py:_recurrent_mutated_genes` selects 193 BRCA genes. The top by recurrence are TP53 (basal/triple-neg), PIK3CA (luminal), GATA3 (luminal lineage), CDH1 (invasive lobular), and TTN, the textbook large-protein passenger that recurs by length alone.

3.4 Q4: Methylation breaks single-node tools

Methylation at 444 M rows / 5.5 GB compressed (~30 GB decompressed) is the only modality that demands streaming; the other five fit comfortably in memory.

The need to handle the methylation parquet in DuckDB streams and Polars lazy collections is what makes the 6-view merge and downstream analyses possible on a single 7.6 GB node.

3.5 Q5: Does the merged feature space preserve modality-specific signal?

Yes. As discussed in Q8 below, MOFA+ identifies two **methylation-only factors** (factors 3 and 7) with no loadings on any other view. The merge does not average modalities together; it preserves modality-distinct signal that downstream factor analysis can disentangle.

4 Supervised: XGBoost on PAM50

4.1 Q6: Multi-omic XGBoost predicts PAM50 well

5-fold-CV accuracy is 0.867 ± 0.009 with macro-F1 0.798 ± 0.027 on the 5-class PAM50 problem.

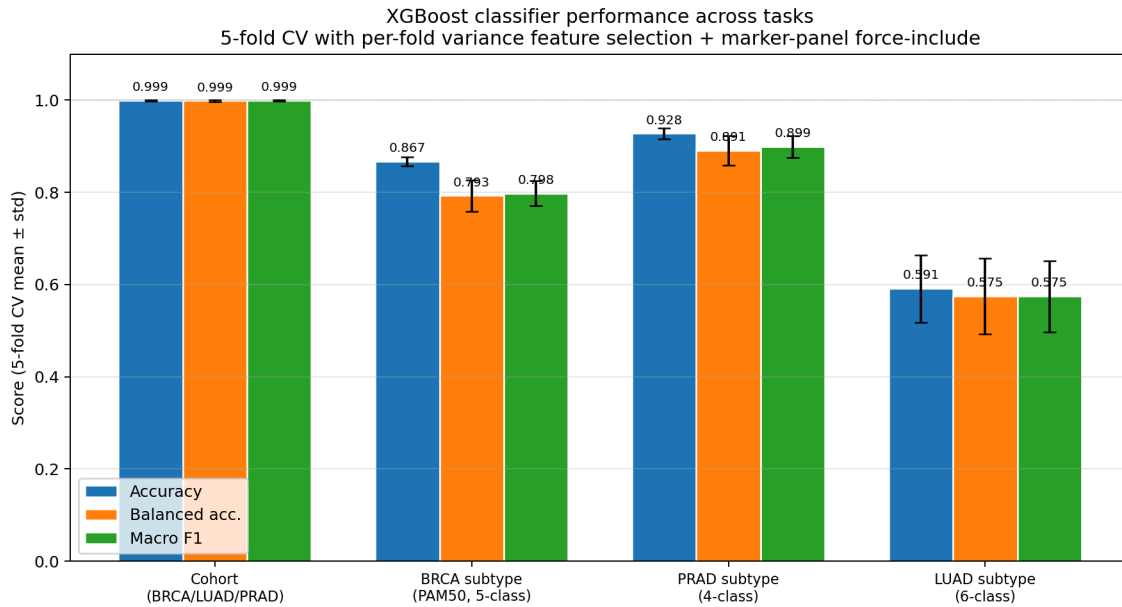


Figure 2: XGBoost performance on the BRCA PAM50 task and on cross-cohort sanity tasks. BRCA-PAM50 reaches 0.87 5-fold-CV accuracy. The 3-cohort separation task is saturating. LUAD subtypes are harder due to the 6-class $\times$ small-cohort combination (Appendix B).

The pipeline is 5-fold stratified XGBoost with per-fold variance re-ranking (top 500 features) plus **force-include** of PAM50 panel genes, so the published markers get a fair shot at appearing in SHAP rankings even if a fold’s variance ranking would have dropped them.

4.2 Q7: Top SHAP features recover the PAM50 panel

17 of 20 PAM50 panel genes appear in the model’s top-50 features by mean $|\text{SHAP}|$, and **9 of 20 in the top-10**. This is strong external validation: the model’s notion of “what matters for PAM50” matches the panel built on independent intrinsic-subtype biology.

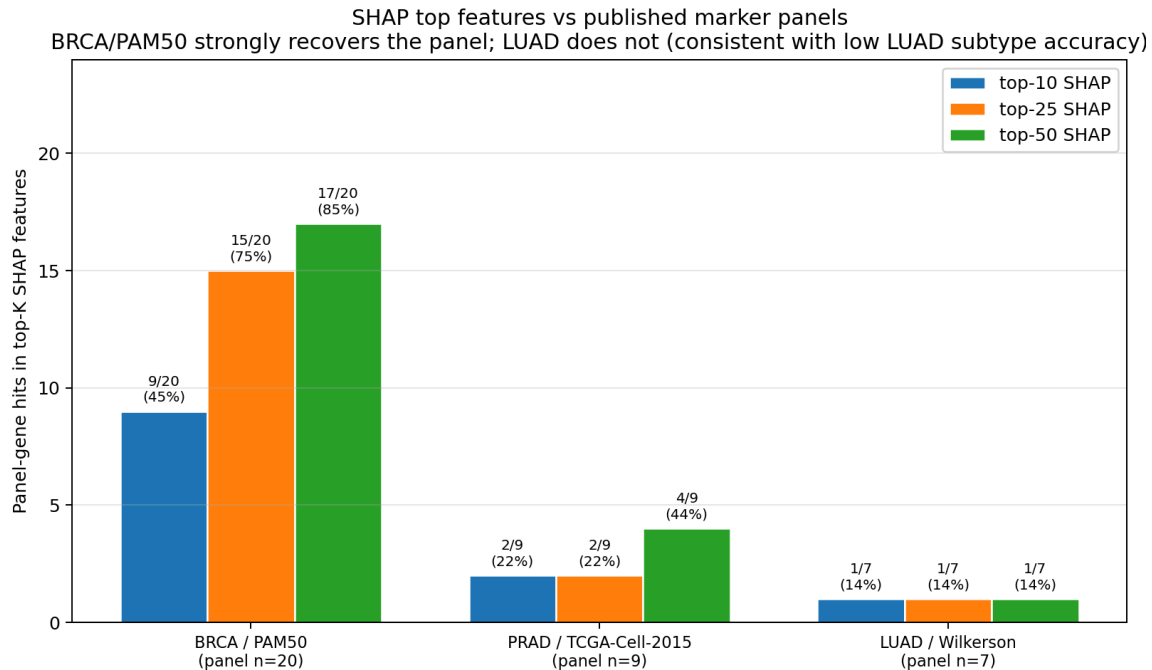


Figure 3: SHAP-vs-panel hits per cohort and rank window (top-10, top-25, top-50). BRCA / PAM50 strongly recovers the panel; LUAD does not, consistent with LUAD’s lower subtype accuracy (Appendix B).

Per-class SHAP plots (data/classification_pipeline_brca/plots/shap_per_class/) show subtype-specific drivers consistent with published intrinsic biology: **Basal** driven by FOXA1↓ + FOXC1↑, **Her2** by ERBB2 ranked #1, **LumA** by AURKA / CDC20 / ESR1 / BCL2.

5 Unsupervised: MOFA+ on Six Modalities

5.1 Q8: MOFA+ identifies a multi-omic dominant factor

MOFA+ converges to **14 active factors** (ARD-pruned from 15), and **factor 1 is the only factor with substantial loadings on all six views simultaneously**. Its top loadings recover textbook luminal-vs-basal biology.

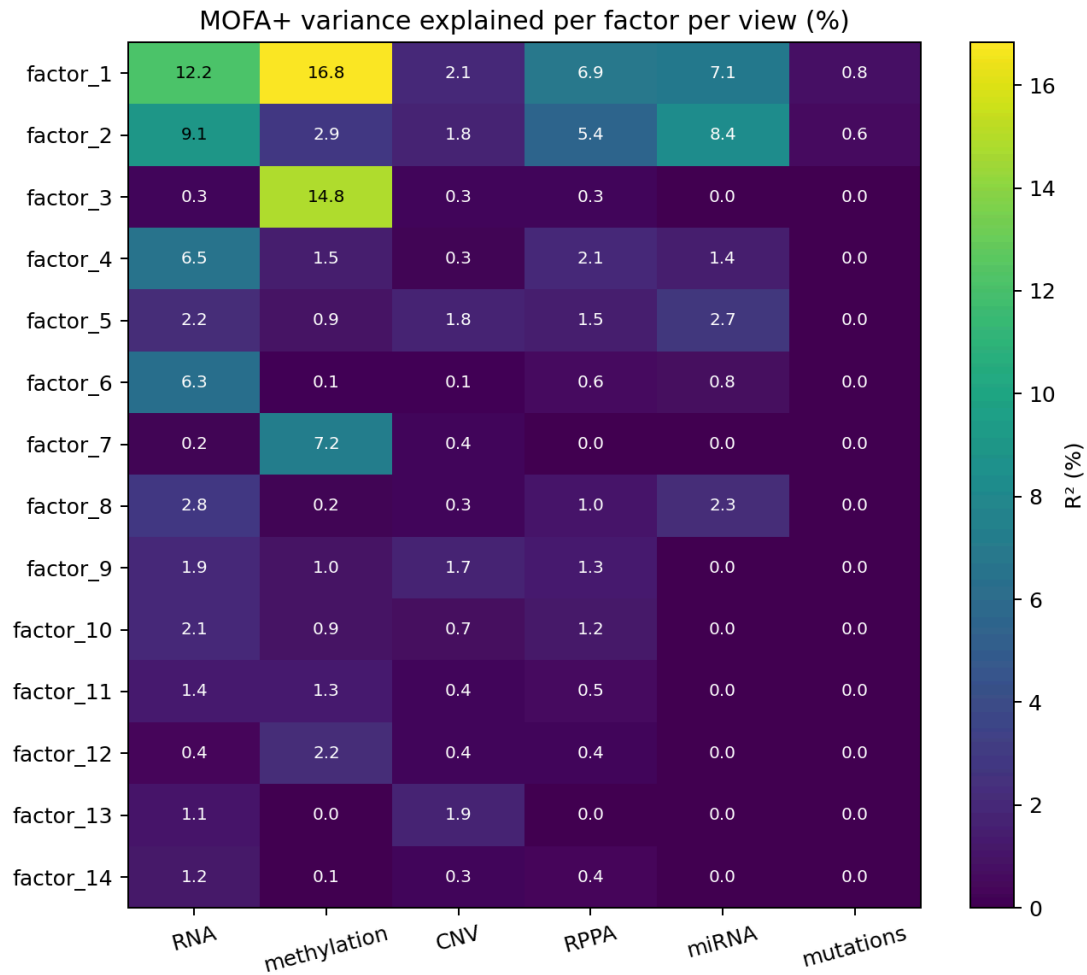


Figure 4: Variance explained per factor per view (%). Factor 1 is multi-omic across all six views (basal-vs-luminal axis); factor 2 captures a within-luminal proliferation gradient; factors 3 and 7 are methylation-specific.

Factor 1 R^2 across views: 12.2% RNA, 16.8% methylation, 2.1% CNV, 6.9% RPPA, 7.1% miRNA, 0.81% mutations. Top loadings on factor 1:

- **RPPA**: ESR1 (ERALPHA), GATA3, AR, PR. The canonical luminal hormone-receptor signature.
- **Mutations**: GATA3 and PIK3CA on the positive side; **TP53 negative**. This is the basal-vs-luminal driver-mutation split documented in TCGA Nature 2012.
- **miRNA**: miR-375, a well-known ER+ luminal marker miRNA.

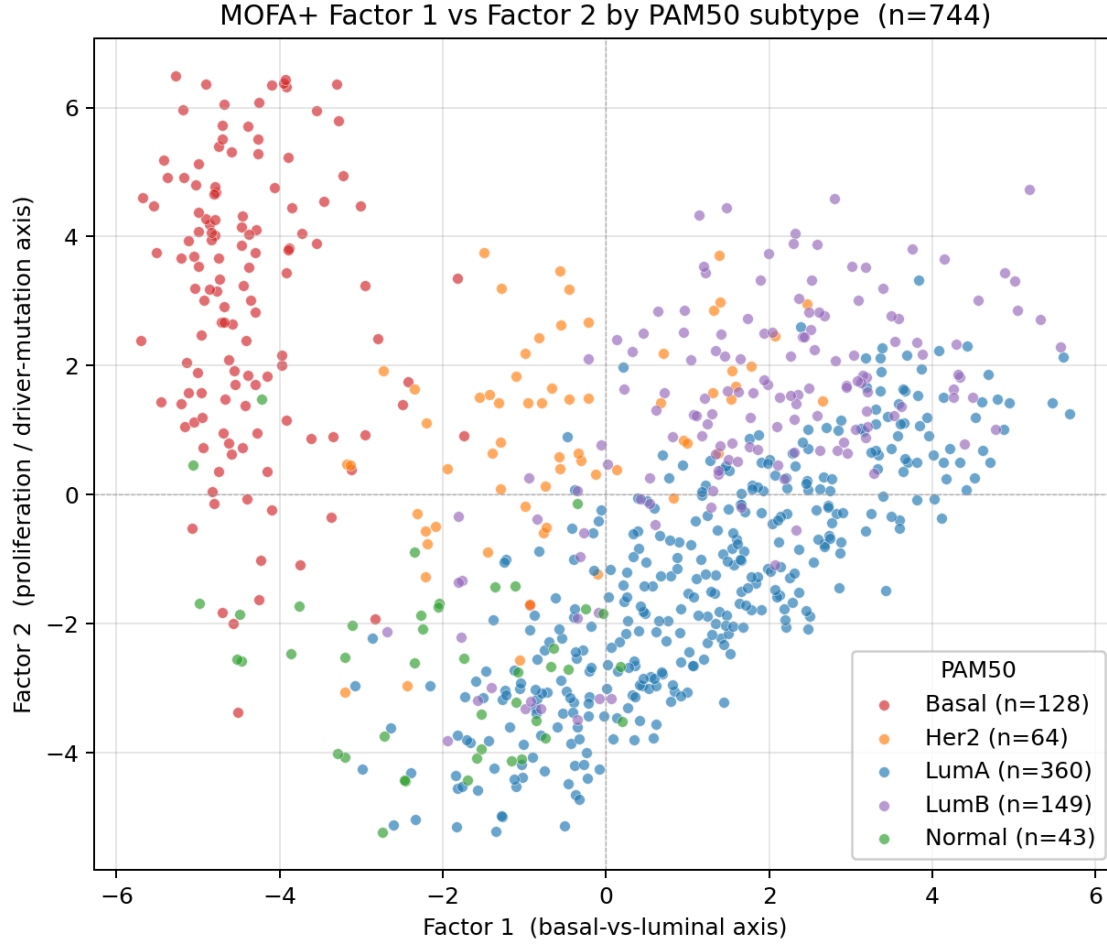


Figure 5: MOFA+ factor 1 vs factor 2, colored by PAM50 subtype. Factor 1 cleanly separates Basal (red) from luminal subtypes; factor 2 captures a within-luminal proliferation gradient.

5.2 Q9: Clustering on factor scores partially recovers PAM50

k-means at PAM50-cardinality ($k=4$) yields $\text{ARI} = 0.268$, $\text{NMI} = 0.328$, below a “good agreement” threshold but with strong substructure. **Cluster 0 is 97% pure Basal-like ($n=97$)** without using any labels, and cluster 2 is 76% pure LumA ($n=278$).

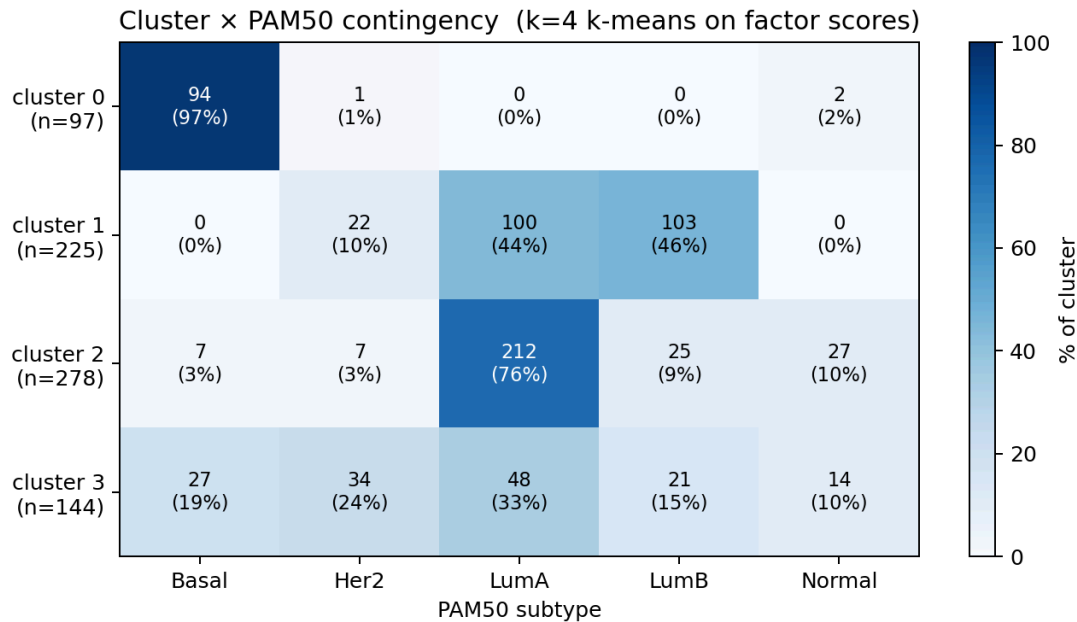


Figure 6: Cluster × PAM50 contingency at k=4. Cluster 0 is 97% pure Basal-like; cluster 2 is 76% LumA. Mixed clusters reflect genuine LumA / LumB / Her2 overlap on multi-omic axes.

The mid-range ARI is informative rather than a failure. PAM50's LumA / LumB / Her2 boundaries are sharper in PAM50's RNA-only definition than in the multi-omic factor space, which integrates CNV, methylation, RPPA, miRNA, and mutations contributing ~60% of total captured variance. Clusters 1 and 3 mixing LumA / LumB / Her2 reflects this.

5.3 Q10: Supervised and unsupervised methods converge on the same axis

Both methods independently surface **basal-vs-luminal** as the dominant signal:

- The supervised XGBoost confusion matrix's hardest *non-Basal* errors are LumA/LumB confusions; Basal is the cleanest class.
- The unsupervised MOFA+ factor 1 cleanly separates Basal from luminal subtypes (Figure 5), and the cleanest cluster (97% Basal-like) is the cluster MOFA+ would have chosen if forced to pick the easiest split first.
- The supervised top SHAP features (FOXA1, FOXC1, ESR1, ERBB2) and the unsupervised factor-1 loadings (ESR1/GATA3/AR/PR proteins, GATA3/PIK3CA vs TP53 mutations) describe the same hormone-receptor axis seen through two windows.

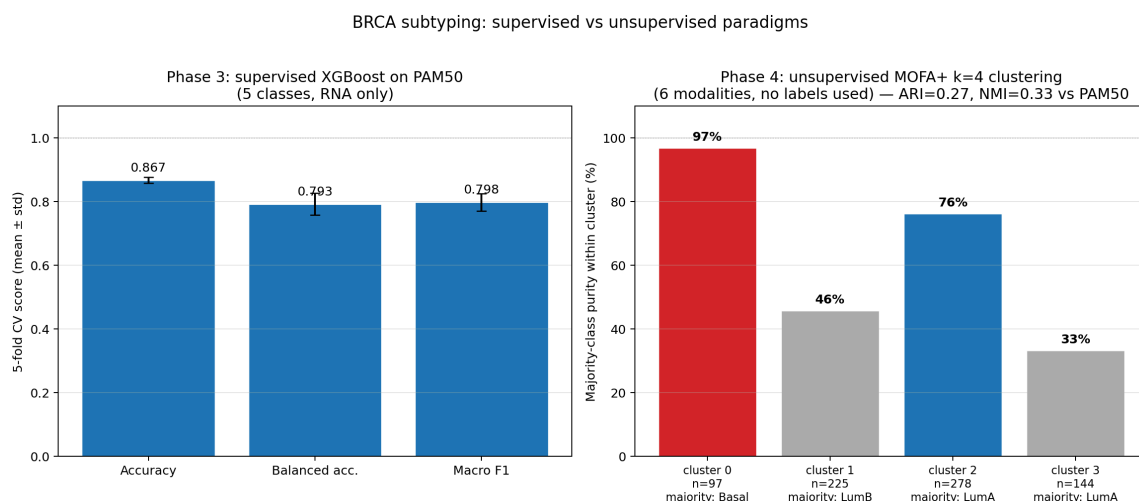


Figure 7: Side-by-side: supervised XGBoost on PAM50 labels (5 classes, 0.87 accuracy) vs unsupervised MOFA+ k=4 clustering (no labels, 97%-pure Basal cluster). Both methods succeed in their own framing and converge on the basal-vs-luminal axis.

With labels, the supervised model treats basal-vs-luminal as one axis among five it has to learn. Without labels, the unsupervised model picks that same axis as its #1 factor.

6 Discussion

6.1 Synthesis

The integrated 6-modality BRCA dataset has a dominant axis that both modeling approaches recover: basal-vs-luminal hormone-receptor biology. Supervised XGBoost confirms PAM50 is learnable (0.87 accuracy) and that its panel-defining genes are also the model’s most-attributing features (17 of 20 in top-50). Unsupervised MOFA+, given no labels, identifies the same axis as its dominant factor and recovers a 97%-pure Basal cluster downstream. The two methods answer different questions (“*can we predict labels?*” vs “*what structure does the data have?*”) and arrive at the same biological story.

6.2 Limitations

- **No external validation cohort.** Metabarc BRCA was scoped but not implemented; cross-platform harmonization (microarray to RNA-seq) is the principal obstacle.
- **Mid-range unsupervised ARI** (0.27–0.31) is below a high-agreement threshold but reflects genuine biology. Multi-omic factors capture cell-of-origin and methylation signal that PAM50 does not see. A fairer comparator would be an iCluster-style integrative subtype label.
- **Spark cluster scaffolded but not exercised.** All science ran on local DuckDB and Polars; the t3 instance was sufficient given memory discipline.

6.3 Future work

Consensus clustering on MOFA+ factors (bootstrap resampling), Similarity Network Fusion [2] as an independent multi-omic comparator, external validation on Metabric, and Spark-cluster-driven pan-cancer scaling beyond the three cohorts.

7 References

8 Appendix

8.1 A. Methods detail

Per-view feature pre-selection. RNA-seq and methylation: top 5,000 features by variance, gated on $\geq 80\%$ **non-null patient coverage** (the gate was critical for methylation; see Q2). RPPA, miRNA, CNV: kept all features. Mutations: kept genes mutated in $\geq 2\%$ **of cohort** (193 BRCA genes). Wide pivot then inner-join on `patient_id` produced the $744 \times 12,612$ multi-omic table.

Supervised classifier. XGBoost.XGBClassifier with 5-fold stratified CV. Per fold: re-rank features by variance and keep top 500, plus **force-include the PAM50 panel** so panel genes get a fair shot at appearing in SHAP rankings even when a fold’s variance would drop them. SHAP attribution via TreeExplainer; mean |SHAP| across folds defines the ranking compared against the panel.

MOFA+. Six views with five Gaussian likelihoods plus Bernoulli for mutations. Settings: 15 requested factors with ARD pruning, `convergence=medium`, `drop_factor_threshold=0.01`. Trained ~16 minutes over ~200 iterations on a single CPU; pruned to 14 active factors. Downstream k-means swept $k=2..10$; ARI and NMI computed against PAM50 at $k=4$.

8.2 B. Cross-cohort sanity

To confirm the pipeline isn’t BRCA-specific, we ran the same XGBoost + SHAP pipeline on TCGA-LUAD (3 modalities, ~230 patients) and TCGA-PRAD (3 modalities). Results:

Task		Accuracy	Macro F1	Panel hits in top-50
3-cohort	BRCA/ LUAD/PRAD	0.999 ± 0.001	0.999 ± 0.001	n/a
BRCA	/ PAM50 (5-class)	0.867 ± 0.009	0.798 ± 0.027	17/20 (PAM50)
PRAD	subtype (4-class)	0.928 ± 0.012	0.899 ± 0.023	by-design (label leakage on ERG/ETV1)
LUAD	subtype (6-class)	0.591 ± 0.073	0.575 ± 0.077	1/7 (Wilkerson)

LUAD’s low subtype accuracy reflects the 6-class $\times$ ~230-patient combination rather than a pipeline bug, and SHAP correspondingly fails to recover the Wilkerson panel. The pipeline scales to LUAD/PRAD but those results are not the focus of this report.

8.3 C. Reproducibility

All code: https://github.com/acarlisle8/cancer_genome_atlas_research.

Stage		Command	Wall time
Ingest		<code>uv run python code/run_ingest.py</code>	~30–60 min
Merge (6-view BRCA)		<code>uv run python code/run_merge_6view.py</code>	~8 min
Supervised + SHAP		<code>uv run python code/run_classification_pipeline.py</code>	~10 min
MOFA+ training		<code>uv run python code/run_mofa.py --cohort BRCA</code>	~16 min
MOFA+ analysis		<code>uv run python code/analyze_mofa.py --model-dir data/mofa_BRCA</code>	<1 min
Render report		<code>quarto render final_report.qmd</code>	<30 s

Compute: AWS EC2 t3 (2 vCPU, 7.6 GB, no swap). Python 3.13 + uv.

Bibliography

- [1] J. S. Parker *et al.*, “Supervised risk predictor of breast cancer based on intrinsic subtypes,” *Journal of Clinical Oncology*, vol. 27, no. 8, pp. 1160–1167, 2009, doi: 10.1200/JCO.2008.18.1370.
- [2] B. Wang *et al.*, “Similarity network fusion for aggregating data types on a genomic scale,” *Nature Methods*, vol. 11, no. 3, pp. 333–337, 2014, doi: 10.1038/nmeth.2810.